

Statement of Work (SOW)

Procurement of Telemetry Interface Modules

1. Objective / Requirements

The SCHlieren, Airborne Measurements, & Range Operations for QueSST (SCHAMROQ) Project at NASA Armstrong Flight Research Center (AFRC) requires the procurement of telemetry interface modules used to support aircraft instrumentation and data acquisition systems.

The required hardware shall be electrically, mechanically, and functionally compatible with the existing **Telemetry Technology Corporation (TTC) TTC's EDAU-20XX and CDAU-20XX systems** currently deployed on AFRC aircraft. The modules shall integrate directly with the installed telemetry chassis and shall not require modification to existing telemetry hardware, firmware, wiring harnesses, telemetry configuration files, or ground telemetry processing systems.

Required hardware:

- Part Number: BLS-148-3
 - Discrete Input Interface Module –
 - Quantity: 2
- Part Number: BIM-232D-4
 - Serial Communication Interface Module –
 - Quantity: 2

No alternates are approved for these parts.

2. Quality Requirements

The contractor shall provide hardware that meets the following requirements:

- Hardware shall be new, unused, and in original manufacturer condition.
- Refurbished, reconditioned, remanufactured, repaired, or surplus hardware shall not be accepted.
- Hardware shall be production units and shall not be prototype, engineering development, or qualification test units.
- Hardware shall include traceability to the Original Equipment Manufacturer (OEM).
- Each delivered item shall include a Manufacturer Certificate of Conformance verifying compliance with the specified part number and requirements.
- Hardware shall comply with NASA Quality Document Q1-B requirements.

3. Warranty Requirements

The contractor shall provide the manufacturer's standard warranty.

The warranty period shall begin upon acceptance of the hardware by NASA.

4. Characteristics, Scope, and Specifications

BLS-148 Interface Module

- Be electrically and mechanically compatible with TTC's EDAU-20XX and CDAU-20XX products.
- Provide discrete signal interface capability for aircraft instrumentation inputs.
- Support monitoring and telemetry mapping of discrete input signals.
- Operate within standard aircraft telemetry electrical and environmental limits.

BIM-232 Interface Module

- Be electrically and mechanically compatible with TTC's EDAU-20XX and CDAU-20XX products.
- Support **RS-232/422/485** serial data ingestion.
- Compatible with RS-232 Honeywell PPT Transducers.
- Support integration of serial data into PCM telemetry data streams.

5. Delivery Requirements – Place of Delivery

NASA Armstrong Flight Research Center
Bldg 4876 Warehouse #6, Lilly Drive
Edwards, CA 93524

6. Freight Delivery Requirements

If freight shipment is required, delivery shall be coordinated at least four business days in advance to ensure proper entry authorization to Edwards Air Force Base.

7. Period of Performance / Delivery Schedule

Hardware shall be delivered no later than **30 weeks** after contract award unless otherwise specified in the contract.